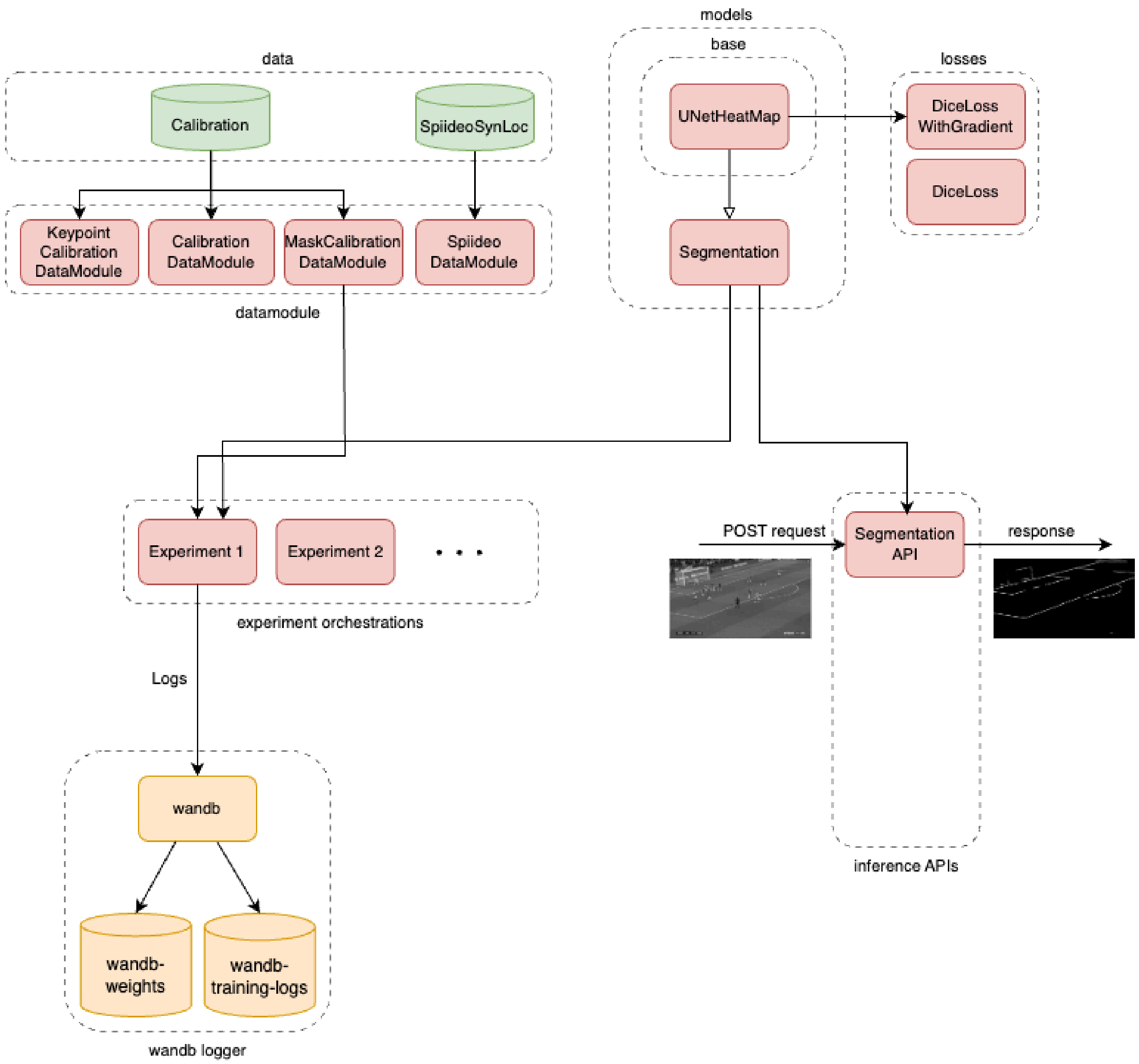
ISC 2, Checkpoint 2

Infrastructure



- ⚠️ Wandb account is currently on Trial and will expire soon.
- Considerations to apply for Wandb for Academics.

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Task Description

- https://github.com/Spiideo/sskit
- https://www.scitepress.org/publishedPapers/2025/131082/pdf/index.html
- Dataset includes 40k+ training / $\sim$ 7k validation/ $\sim$ 9k test images of football matches, synthetically generated from 3D rendering techniques.
  - COCO format: `image`, `annotations`, `categories`.
- Features per image:
  - Calibration parameters:
    - *Camera matrix*: $\boldsymbol{P}_{3\times 4} = [\boldsymbol{R}_{3\times 3} \mid \tilde{\boldsymbol{t}}]$
    - *Industrial distortion model*:
      - Normalised polar coordinates:
        - $(u, v) = \frac{1}{w}\left(x - \frac{w}{2}, y - \frac{h}{2}\right)$
        - $r = \sqrt{u^2 + v^2}$
      - Distortion: $r_{\text{dist}} = p_{\text{dist}}\left(\tan^{-1}(r_{\text{undist}})\right)$
      - Undistortion: $r_{\text{undist}} = \tan(p_{\text{undist}}(r_{\text{dist}}))$
      - p_{dist} and p_{undist} are given polynomials. $p_{\text{dist}}^{-1} \approx p_{\text{undist}}$
- Features per each annotated player:
  - Keypoints
    - 2 keypoints recorded, at the pelvis and at its projection to the ground.
    - Both the world coordinates (3D, in world meters) and the image coordinates (2D, in image pixels) are provided.
  - **Location on the pitch** (in meters)
  - Bounding box
- Task: **determine the pitch location of every player**.

Figure 1: Spiideo synthetic dataset. Each player is annotated by two keypoints: the pelvis and the projection of the pelvis onto the field ground.

Figure 2: Spiideo synthetic dataset, after undistortion.

- Metric: *mAP-LocSim*
  - $\text{LocSym}(\boldsymbol{x}_1, \boldsymbol{x}_2) = \exp\left(\ln 0.05 \cdot \frac{d(\boldsymbol{x}_1, \boldsymbol{x}_2)^2}{\tau^2}\right)$, where $\tau = 1$ m is the tolerance constant –> “target on precise location”

Models

Approach	Drawbacks	Status
• Train a **YOLO** detector to detect regions associating to each player • Determine the keypoints for each player based on the cropped region	• Images are blurry and contain artifacts $\longrightarrow$ harder to detect objects • (Ultralytics) YOLO does not provide a flexible interface for training (no custom dataset, etc.)	-/-
• Segmentation model (**UNet**-based) • Determine the keypoints for each player based on the cropped region • Code: `./models/base/unet_heatmap.py`	• UNet isn’t stable, may produce false negative pixels. • This behaviour can be seen from the `MaskCalibration` model.	**in research**
• **Gaussian Mixture Model** (?) • Treat each player as a Gaussian distribution (heatmap, etc.) • Code: `./pipelines/player_detection/location_detector.py`	-/-	**in research**